

Fast Tsunami Surrogate Modeling with Fourier Neural Operators

Generalization, Resolution Transfer, and Uncertainty Estimation

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Abstract

We study fast neural-operator surrogates for tsunami-like wave propagation in a controlled synthetic setting based on two-dimensional shallow-water simulations. Our pipeline generates diverse bathymetry and source families, and uses the resulting numerical trajectories as reference data for supervised learning. We evaluate a Fourier Neural Operator (FNO) as the primary surrogate model and compare it against convolutional baselines under matched training and evaluation protocols. The experiments focus on three questions: fidelity to the numerical solver, inference speedup, and robustness under distribution shift, including unseen bathymetry families and cross-resolution transfer from coarse to finer grids. Our results are intended to assess whether operator-learning models can provide meaningful acceleration for tsunami surrogate modeling while preserving acceptable predictive accuracy in a research setting. We emphasize that this work targets controlled surrogate modeling rather than operational early-warning deployment, including unseen bathymetry families and cross-resolution transfer from coarse to finer grids, with an additional uncertainty analysis of predictive confidence.

1. Introduction

Physics-based tsunami prediction remains computationally expensive, often requiring large-scale parallel computation when evaluated across many scenarios, parameter sweeps, or inverse problems. This motivates the search for surrogate models that can approximate tsunami propagation far more efficiently while preserving useful predictive accuracy.

In this work, we cast surrogate modeling as an operator-learning problem: given a bathymetry field and an initial disturbance, the model predicts the wave-height field over future timesteps. Fourier Neural Operators (FNOs) are a compelling choice because they learn mappings between functions and have shown strong performance on PDE-driven tasks.

Our aim is not to present an operational warning system. Instead, we establish a clean research benchmark that lets us ask a sharper question: can a neural operator deliver a meaningful speed--accuracy trade-off for tsunami-like wave propagation under controlled synthetic conditions?

The main contributions of this study are:

1. A realistic, reproducible synthetic tsunami data pipeline built from a shallow-water solver,
2. a neural-operator benchmark centered on FNOs with CNN, U-Net, and ConvLSTM baselines,

3. an evaluation suite covering accuracy, speed, generalization, and uncertainty, and
4. a curriculum-by-resolution experiment that tests whether progressive training improves high-resolution prediction quality.

2. Related works

3. Data generation pipeline

3.1. Synthetic Bathymetry Generation

3.1.1. Objective and Motivation

We model bathymetry procedurally so that each training sample contains a different seafloor geometry rather than a fixed coastal map. This design choice lets us generate large numbers of physically plausible bathymetric fields with controlled variation in slope, roughness, basin structure, and nearshore features, while keeping all samples on a common numerical grid and depth range.

The motivation is twofold. First, tsunami propagation and coastal amplification depend strongly on seafloor structure, so a surrogate model trained on only one or a few bathymetries would risk learning location-specific patterns instead of broadly useful wave dynamics. Second, real bathymetric products are often incomplete, heterogeneous, or expensive to assemble at scale. A procedural generator therefore provides a reproducible way to create diverse training environments without requiring a large curated database of real coastlines.

This diversity is especially important for the Fourier Neural Operator (FNO), which is intended to learn an operator that maps from bathymetry and initial disturbance to future wave fields. To encourage the model to learn transferable wave physics rather than memorizing the spectral signature of a particular domain, we treat bathymetry as a variable input. In this sense, synthetic bathymetry generation is not only a data augmentation step, but a core part of the benchmark design for evaluating generalization.

3.1.2. Domain and sign convention

All synthetic samples are defined on a fixed two-dimensional Cartesian grid of size $n_x \times n_y$. The bathymetry and source generators are configured to use the same spatial resolution as the shallow-water solver, so every field is represented on a shared lattice with spacings Δx and Δy . In the default setup, the procedural bathymetry is first generated on normalized coordinates over the unit square and then passed to the solver on the corresponding discrete grid.

We adopt the convention that bathymetry values are non-positive below sea level, with $b(x, y) = 0$ at mean sea level and $b(x, y) < 0$ in submerged regions. After generation, bathymetric fields are clipped to a prescribed depth range, which in our experiments places the seafloor between a

maximum depth and zero elevation. The corresponding still-water depth is then computed as

$$h_{\text{rest}}(x, y) = \max(-b(x, y) + \eta_{\text{sea}}, 0),$$

where η_{sea} is an optional sea-level offset. This guarantees non-negative water depth even in cells that would otherwise be dry.

The tsunami source field is interpreted as an initial free-surface displacement template and scaled by a randomly sampled source strength. If $\eta_0(x, y)$ denotes this initial surface perturbation, then the initial total water depth supplied to the solver is

$$h_0(x, y) = \max(h_{\text{rest}}(x, y) + \eta_0(x, y), 0).$$

The initial free-surface elevation is recovered as

$$\eta(x, y, 0) = h_0(x, y) + b(x, y).$$

Under this convention, positive η corresponds to an uplift of the water surface and negative η corresponds to a depression. This sign choice is used consistently throughout dataset generation, simulation, and preprocessing.

3.2. Earthquake source

3.3. Dataset generator

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